

EXACT SEMIPRIMES IN ALMOST ALL INTERVALS OF LENGTH $(\log X)^{2.092}$

ABSTRACT. Matomäki and Teräväinen proved that almost every interval $(x, x + (\log x)^{2.1}]$ contains a product of exactly two primes. We show that the exponent 2.1 can be replaced by any fixed

$$c > \frac{3688}{1763} = 2.091888\dots,$$

and in particular by 2.092; quantitatively, almost every such interval contains $\gg (\log x)^{1.092}$ products pq with $(\log x)^{1.091} < p \leq (\log x)^{1.092}$. This beats the Matomäki–Teräväinen exponent 2.1 by more than 0.008, and it also goes below 197/94, the limit of their argument as it stands, and below 67/32, the value obtained by inserting the Guth–Maynard large-value theorem with a single threshold. The proof keeps the Harman-sieve and sparse-mean-value architecture of Matomäki and Teräväinen and changes how the large values of the two type II factors are controlled. Instead of one large-value threshold for both factors, we use the Guth–Maynard large-value estimates with a threshold that depends on the length of the polynomial: the short factor is powered into the range $[X^{5/6}, X)$, where the Guth–Maynard long-polynomial bound allows every exponent below $1/4$, and the long factor of length $X^{1-\theta}$ gets the threshold $\frac{3-8\theta}{10(1-2\theta)}$ (for $1/6 \leq \theta \leq 2/11$; it is $1/4$ for smaller θ). A three-case analysis of the medium large-value bins then closes the two-branch sparse argument for every $a > 1925/1763$, where $P_1 = (\log X)^a$. The whole deduction, from the cited published theorems and five explicitly labelled uniform-in-parameter versions of Matomäki–Teräväinen lemmas, has been checked in Lean 4 with Mathlib.

1. INTRODUCTION

Let

$$E_2 := \{pq : p, q \text{ prime}\}$$

be the set of integers with exactly two prime factors, counted with multiplicity. Unlike integers with *at most* two prime factors, E_2 is subject to the parity problem, and finding its elements in short intervals requires genuinely bilinear (type II) information.

The previous record is due to Matomäki and Teräväinen.

Theorem 1.1 (Matomäki–Teräväinen [6, Theorem 1.1]). *There exist constants $c_0 > 0$ and $\delta > 0$ such that, for all $X \geq 3$, all but $O(X/(\log X)^\delta)$ integers $x \in [2, X]$ satisfy*

$$\#\{p_1 p_2 \in (x, x + (\log x)^{2.1}] : (\log x)^{1.09} < p_1 \leq (\log x)^{1.1}\} \geq c_0 (\log x)^{1.1}.$$

Our main result shortens the interval.

Theorem 1.2 (Exponent 2.092). *There are constants $c_0, \delta > 0$ such that, for all sufficiently large X , all but $O(X/(\log X)^\delta)$ integers $x \leq X$ satisfy*

$$\#\{pq \in (x, x + (\log x)^{2.092}] : p, q \text{ prime}, (\log x)^{1.091} < p \leq (\log x)^{1.092}\} \geq c_0 (\log x)^{1.092}.$$

In particular, almost every interval $(x, x + (\log x)^{2.092}]$ contains an element of E_2 .

The exponent 2.092 is a convenient decimal; the method gives the following.

Theorem 1.3 (Parameter form). *For every fixed*

$$c > \frac{3688}{1763} = 1 + \frac{1925}{1763} = 2.091888\dots,$$

Date: 3 October 2026. Research draft; not peer reviewed.

almost every interval $(x, x + (\log x)^c]$ contains an element of E_2 ; the number of exceptional integers $x \leq X$ is $O_c(X/(\log X)^{\delta_c})$ for some $\delta_c > 0$. If moreover $c \leq 21/10$ and $\Delta > 0$ satisfies $c - 1 - \Delta > 1925/1763$, then, outside an exceptional set of the same type, the interval $(x, x + (\log x)^c]$ contains $\gg_{c,\Delta} (\log x)^{c-1}$ products pq with $(\log x)^{c-1-\Delta} < p \leq (\log x)^{c-1-\Delta/2}$.

How this compares with Matomäki–Teräväinen. Theorem 1.2 beats the published exponent 2.1 of [6, Theorem 1.1]: the interval length drops by the factor

$$\frac{(\log x)^{2.1}}{(\log x)^{2.092}} = (\log x)^{0.008} \rightarrow \infty,$$

and the threshold $3688/1763$ is below 2.1 by

$$\frac{21}{10} - \frac{3688}{1763} = \frac{143}{17630} > 0.0081.$$

The improvement is not an artefact of the decimals chosen in [6]. Their argument, optimized as it stands (Jutila’s large-value threshold $49/206$ in [6, Lemma 4.1] and the two sparse branches of [6, Section 5.3]), reaches $c > 1 + 103/94 = 197/94$ and no further; see the last display of [6, Section 5.3]. Replacing Jutila’s estimate by the Guth–Maynard large-value theorem with a single threshold $17/70$ for both type II factors gives $c > 67/32$. The present paper goes below both:

Argument	large-value input	admissible c
[6, Thm. 1.1], as printed	Jutila, $49/206$	$c = 2.1$
[6], optimized	Jutila, $49/206$	$c > 197/94 = 2.09574\dots$
single threshold	Guth–Maynard, $17/70$	$c > 67/32 = 2.09375$
this paper	Guth–Maynard, by length	$c > 3688/1763 = 2.09188\dots$

In the language of [6]: a pair of polynomial lengths $M_1 = X^\theta$, $M_2 = X^{1-\theta}$ with $\theta \leq 2/11$ never needs the worst threshold $17/70$ on *both* factors. The short factor can always be powered into $[X^{5/6}, X)$, and the long factor needs $17/70$ only when $\theta = 2/11$ exactly. Section 6 turns this observation into the bound $a > 1925/1763$.

Formal verification. The deduction of Theorems 1.2 and 1.3 has been formalized in Lean 4 with Mathlib (Section 8). The Lean theorems take two explicit hypothesis bundles: published theorems transcribed in their printed ranges (Guth–Maynard, Heath-Brown, Hildebrand–Tenenbaum, Matomäki–Teräväinen, Iwaniec–Kowalski, $\dots$), and five labelled uniform-in-parameter versions of Matomäki–Radziwiłł, Teräväinen and Matomäki–Teräväinen lemmas that are printed only for $c = 2.1$. All of the new mathematics of this paper (Lemma 4.3, Proposition 6.1 and everything after) is proved in Lean, not assumed.

Status and provenance. This is a research draft and has not been refereed. An earlier draft, generated with ChatGPT (OpenAI), obtained $c > 67/32$ with a single threshold $17/70$; the length-dependent refinement, the present write-up and the Lean formalization of both were produced with Claude (Anthropic). Readers should check the argument independently before relying on it.

2. NOTATION

All logarithms are natural; p, q denote primes and $n \sim N$ means $N < n \leq 2N$. We write $A \ll B$ or $A = O(B)$ if $|A| \leq CB$ for a constant C depending only on the displayed subscripts and on fixed parameters. The parameters are chosen in the following order: first the gap $\nu > 0$ between a and $1925/1763$; then the analytic parameter $\varepsilon > 0$, small in terms of ν and of the absolute constant C_1 of Proposition 5.1; finally X is taken large in terms of everything else.

A sequence (a_n) is (A_0, B) -divisor bounded if $|a_n| \leq A_0 d(n)^B$ on its support, with $A_0 \geq 1$, $B \geq 0$ independent of the principal scales. If a is (A_0, B) -divisor bounded, then the coefficients of the ℓ -fold Dirichlet convolution of a restricted to a dyadic range are $(A_0^\ell, B\ell + \ell - 1)$ -divisor bounded: the first $\ell - 1$ factors of an ordered factorization of n are divisors of n .

For a Dirichlet polynomial A , a height T and a level $V > 0$ we write

$$\mathcal{L}(A; T, V) = \{t \in [0, T] : |A(1 + it)| > V\}.$$

3. THE FRAMEWORK OF MATOMÄKI AND TERÄVÄINEN

We recall the part of [6] that we use, with the parameters left free. Let $c \in (2, 21/10]$, $h = (\log X)^c$, and let $P_1 = (\log X)^a$ with $1 < a \leq c - 1 \leq 11/10$. Harman's sieve with $z = X^{2/11}$ provides a minorant $\rho^-(n) \leq \mathbf{1}_{\mathbb{P}}(n)$ [6, (2.3)] with positive average on long intervals [6, Theorem 2.1(ii)]. By Perron's formula [6, Lemma 3.1] (after [5, Lemma 14] and [7, Lemma 1]), the variance of $\sum_{x < pn \leq x+h, p \sim P_1} \rho^-(n)$ is controlled by mean values

$$\int_{X^{1/1000}}^T |P_1(1 + it)|^2 |P(1 + it)|^2 dt, \quad P_1(s) = \sum_{p \sim P_1} p^{-s}, \quad P(s) = \sum_{X/(2P_1) < n \leq 4X/P_1} \frac{\rho^-(n)}{n^s}.$$

On the set

$$(1) \quad \mathcal{U} = \{t \in [X^{1/1000}, X] : |P_1(1 + it)| \geq P_1^{-\varepsilon/10}\}$$

the polynomial P is decomposed by [6, Proposition 2.2] into $O(\exp((\log \log X)^5))$ type I, type I/II and type II convolutions and a negligible remainder. The type I and type I/II components are estimated in [6, Sections 5.1–5.2]; their only dependence on a is through the power savings $X^{-(1-1/a)/20+O(\varepsilon)}$ and $X^{-\varepsilon(1-1/a)/4+\varepsilon/100+O(\varepsilon^2)}$, which remain power savings for $a \geq 1925/1763$ because

$$\frac{1 - 1/a}{4} \geq \frac{1 - 1763/1925}{4} = \frac{81}{3850} > \frac{1}{100}.$$

A type II component is $F = M_1 M_2$, where

$$M_1(s) = \sum_{m \sim M_1} \frac{\alpha_m}{m^s}, \quad M_2(s) = \sum_{n \sim M_2} \frac{\beta_n}{n^s}, \quad M_1 M_2 \asymp \frac{X}{P_1}, \quad X^{\varepsilon/2} \leq M_1 \leq X^{2/11},$$

with divisor-bounded coefficients and α of the structured form [6, (2.11)]. We write $\theta = \log M_1 / \log X \in [\varepsilon/2, 2/11]$; then $M_2 = X^{1-\theta-o(1)}$. The target is

$$(2) \quad \int_{\mathcal{U}} |P_1(1 + it)|^2 |M_1(1 + it) M_2(1 + it)|^2 dt \ll \exp(-(\log \log X)^6).$$

As in [6, Section 5.3], $\mathcal{U}$ is split into the set $\mathcal{U}_1$ where $|M_j(1 + it)| \geq M_j^{-\sigma_*}$ for some j , the set $\mathcal{U}_3$ where some $|M_j(1 + it)| < X^{-1}$, and the rest $\mathcal{U}_2$, which is cut into $O((\log X)^2)$ bins

$$(3) \quad \mathcal{V}_{\sigma_1, \sigma_2} = \{t \in \mathcal{U}_2 : M_j^{-\sigma_j} < |M_j(1 + it)| \leq e M_j^{-\sigma_j} \ (j = 1, 2)\}, \quad \sigma_1, \sigma_2 > \sigma_*.$$

The set $\mathcal{U}_1$ is handled by the type II estimate (Proposition 5.1) together with the Vinogradov–Korobov bound [6, (5.6)] $\sup_{t \in \mathcal{U}} |M_1(1 + it)| \ll \exp(-2(\log X)^{1/10})$; $\mathcal{U}_3$ is trivial. Everything therefore reduces to the bin estimate

$$(4) \quad |\mathcal{V}_{\sigma_1, \sigma_2}| \ll M_1^{2\sigma_1} M_2^{2\sigma_2} X^{-\varepsilon^2/4},$$

uniformly in $\sigma_1, \sigma_2 > \sigma_*$ and in the component. This is where the exponent a , and hence c , is decided.

4. LENGTH-DEPENDENT LARGE-VALUE THRESHOLDS

We use two results of Guth and Maynard.

Theorem 4.1 (Guth–Maynard [1, Theorem 1.1]). *Let $|b_n| \leq 1$ and let $t_1, \dots, t_R \in [0, T]$ be one-spaced with $|\sum_{N < n \leq 2N} b_n n^{it_r}| \geq V$ for all r . Then*

$$R \leq T^{o(1)} (N^2 V^{-2} + N^{18/5} V^{-4} + T N^{12/5} V^{-4}).$$

Theorem 4.2 (Guth–Maynard [1, Proposition 12.1]). *Under the hypotheses of Theorem 4.1, if $T^{5/6} \leq N \leq T$ and $V = N^\varrho$ with $\varrho \geq 7/10$, then*

$$R \leq T^{o(1)} (N^{2-2\varrho} + T^{1/2} N^{3-4\varrho} + T^{(30\varrho-21)/5} N^{(46-60\varrho)/5}).$$

For $9/11 \leq \eta \leq 1$ put

$$s(\eta) = \begin{cases} \frac{8\eta-5}{10(2\eta-1)}, & 9/11 \leq \eta \leq 5/6, \\ \frac{1}{4}, & 5/6 \leq \eta \leq 1, \end{cases} \quad s\left(\frac{9}{11}\right) = \frac{17}{70}, \quad s\left(\frac{5}{6}\right) = \frac{1}{4}.$$

The function s is increasing and continuous. The value $17/70$ used in the single-threshold argument is its minimum, attained only at the shortest length $T^{9/11}$.

Lemma 4.3 (Refined density estimate). *Let $0 < \varepsilon \leq 1/1000$, $A_0 \geq 1$, $B \geq 0$. There are C, T_0 such that the following holds for $T \geq T_0$. Let $\eta_0 \in [9/11, 5/6]$, let N be an integer with $T^{\eta_0-10\varepsilon} \leq N \leq T^{1-\varepsilon/10}$, let $A(s) = \sum_{n \sim N} a_n n^{-s}$ with (A_0, B) -divisor bounded coefficients, and let σ satisfy*

$$(5) \quad 10\varepsilon \leq \sigma \leq \frac{1}{4} - 20\varepsilon, \quad \sigma(4\eta_0 - 2) + 20\varepsilon \leq \frac{8\eta_0}{5} - 1.$$

Then

$$|\mathcal{L}(A; T, N^{-\sigma})| \leq C T^{2\sigma - \varepsilon^2/2}.$$

The second condition in (5) says $\sigma \leq s(\eta_0) - O(\varepsilon)$ when $\eta_0 < 5/6$; at $\eta_0 = 5/6$ it is implied by the first.

Proof. Write $N = T^\eta$. Normalize as in [6, Lemma 4.1]: with a fixed $0 < \varpi \leq \varepsilon^2/1000$, choose $D = D(B, \varpi)$ with $d(n)^B \leq Dn^\varpi$ and put $b_n = \frac{N}{n} a_n / (A_0 D 2^\varpi N^\varpi)$, so that $|b_n| \leq 1$ and $|A(1+it)| > N^{-\sigma}$ implies $|\sum b_n n^{-it}| \geq N^{1-\sigma-2\varpi}$ for large T . Pass from measure to one-spaced sets by the standard local discretization, and reflect $t \mapsto T - t$ if needed.

Case $\eta \leq 5/6$. Theorem 4.1 with $V = N^{1-\sigma-2\varpi}$ gives

$$R \leq T^{o(1)} (N^{2\sigma+4\varpi} + N^{4\sigma-2/5+8\varpi} + T N^{4\sigma-8/5+8\varpi}).$$

Compare each exponent of T with 2σ . The first is $2\sigma\eta \leq 2\sigma - 2\sigma\varepsilon/10 \leq 2\sigma - 2\varepsilon^2$. For the second, since $4\eta - 2 > 0$ and $\sigma \leq 1/4 - 20\varepsilon$,

$$\eta(4\sigma - \frac{2}{5}) - 2\sigma = \sigma(4\eta - 2) - \frac{2\eta}{5} \leq \frac{3\eta}{5} - \frac{1}{2} - 20\varepsilon(4\eta - 2) < -20\varepsilon,$$

using $\eta \leq 5/6$. The third exponent $1 + \eta(4\sigma - 8/5)$ is decreasing in η , so it is at most its value at $\eta = \eta_0 - 10\varepsilon$:

$$1 + \eta_0(4\sigma - \frac{8}{5}) + 10\varepsilon(\frac{8}{5} - 4\sigma) \leq 2\sigma - 20\varepsilon + 16\varepsilon = 2\sigma - 4\varepsilon$$

by (5).

Case $\eta \geq 5/6$. Theorem 4.2 with $\varrho = 1 - \sigma - 2\varpi \geq 7/10$ gives the exponents $2\sigma\eta + O(\varpi)$,

$$\frac{1}{2} + \eta(4\sigma - 1) + O(\varpi) = 2\sigma - (2\eta - 1)(\frac{1}{2} - 2\sigma) + O(\varpi) \leq 2\sigma - \frac{80}{3}\varepsilon + O(\varpi),$$

and

$$\begin{aligned} \frac{9-30\sigma}{5} + \eta \frac{60\sigma-14}{5} + O(\varpi) &= 2\sigma + \frac{1}{5}(9 - 14\eta - \sigma(40 - 60\eta)) + O(\varpi) \\ &\leq 2\sigma - \frac{1}{5}(1 - \eta + 20\varepsilon(60\eta - 40)) + O(\varpi), \end{aligned}$$

where we used $\sigma \leq 1/4 - 20\varepsilon$ and $60\eta - 40 > 0$. All savings are at least $2\varepsilon^2 - O(\varpi)$, and the $T^{o(1)}$ and normalization losses are absorbed for large T . $\square$

Remark 4.4. Lemma 4.3 is exactly what the two Guth–Maynard bounds give; nothing beyond [1] is used. The single-threshold argument uses only the value $s(9/11) = 17/70$ and applies it to both type II factors, whatever their lengths.

5. THE TYPE II ESTIMATE WITH THRESHOLD 17/70

Lemma 4.3 with $\eta_0 = 9/11$ and $\sigma \leq 17/70 - O(\varepsilon)$ covers every length in $[T^{9/11-10\varepsilon}, T^{1-\varepsilon/10}]$. Inserting it into the proof of [6, Proposition 4.2] in place of [6, Lemma 4.1] gives the following.

Proposition 5.1 (Type II, threshold 17/70). *There is an absolute constant C_1 such that the following holds for $\varepsilon > 0$ small. Let M_1, M_2 be Dirichlet polynomials of lengths M_1, M_2 with divisor-bounded coefficients, $(\log T)^{-L} \leq M_1 M_2 / T \leq (\log T)^L$ and $T^{\varepsilon/5} \leq M_1 \leq T^{2/11+\varepsilon}$, and put $\sigma_* = 17/70 - C_1\varepsilon$. If $\mathcal{U}' \subset [0, T]$ is measurable and on $\mathcal{U}'$ one has $|M_1(1+it)| \geq M_1^{-\sigma_*}$ or $|M_2(1+it)| \geq M_2^{-\sigma_*}$, then*

$$\int_{\mathcal{U}'} |M_1(1+it)M_2(1+it)|^2 dt \ll T^{-\kappa_1} + (\log T)^{C_2} \sup_{t \in \mathcal{U}'} |M_1(1+it)|^2$$

for some $\kappa_1 = \kappa_1(\varepsilon, B) > 0$ and $C_2 = C_2(\varepsilon, A_0, B, L)$.

Proof. As in [6, Proposition 4.2]: the large-value parts $\mathcal{R}_1, \mathcal{R}_3$ are unchanged; in the middle range one powers M_1 into $[T^{5/6-\varepsilon}, T^{1-\varepsilon/2}]$ when $\tau_1 \leq \tau_2$, or applies the density bound to M_2 (of length in $[T^{9/11-2\varepsilon}, T^{1-\varepsilon/6}]$) when $\tau_2 < \tau_1$, now with Lemma 4.3 at $\eta_0 = 9/11$ instead of Jutila's estimate. $\square$

This is the large-value set $\mathcal{U}_1$ of Section 3. The new work is in the bins $\mathcal{V}_{\sigma_1, \sigma_2}$, where both $\sigma_j > \sigma_*$.

6. THE MEDIUM BINS

Fix $\nu > 0$ and let

$$\frac{1925}{1763} + \nu \leq a \leq \frac{11}{10}, \quad u = 1 - \frac{1}{a}.$$

Then $u \geq 162/1925 + \frac{10}{11} \cdot \frac{1763}{1925}\nu$. Consider one bin $\mathcal{V} = \mathcal{V}_{\sigma_1, \sigma_2}$ with $\sigma_1, \sigma_2 > \sigma_*$, and write

$$\theta = \frac{\log M_1}{\log X} \in \left[\frac{\varepsilon}{2}, \frac{2}{11}\right], \quad \phi = \frac{\log M_2}{\log X}, \quad 1 - \theta - o(1) \leq \phi \leq 1 - \theta.$$

6.1. The sparse branches. We recall [6, Section 5.3]. With $k = \lfloor \log X / \log(2P_1) \rfloor$ and $M = P_1^k = X^{1-o(1)}$, the sparse polynomial $Q = P_1^k / k!$ has $|Q(1+it)| \geq M^{-\varepsilon/10-1/a+o(1)}$ on $\mathcal{U}$ and support of size $M^{1-1/a+o(1)}$ by Hildebrand–Tenenbaum [3]. Heath-Brown's sparse mean value theorem [2, Theorem 4], applied to Q and M_2 , proves (4) unless

$$(6) \quad \theta(1 - 2\sigma_1) \geq u - O(\varepsilon) \quad (\text{the first branch fails}).$$

Applied to Q and M_1^5 (the first term of [6, Lemma 3.4] is absorbed because $M_1^5 \leq X^{10/11} \leq X^{1/a}$), it proves (4) whenever

$$(7) \quad K(\theta, \sigma_1, \sigma_2) := (5 - 8\sigma_1)\theta + 2\sigma_2(1 - \theta) > \frac{1}{a} + O(\varepsilon).$$

Note the identity

$$(8) \quad K(\theta, \sigma_1, \sigma_2) = 4\theta(1 - 2\sigma_1) + \theta(1 - 2\sigma_2) + 2\sigma_2.$$

Under (6) we have $\theta \geq 1/7$, since $1 - 2\sigma_1 < 1 - 2\sigma_* \leq 18/35 + O(\varepsilon)$ and $u \geq 162/1925$.

6.2. The density cases. From now on assume that the first branch fails, so that (6) holds and $1/7 \leq \theta \leq 2/11$.

- (a) $\sigma_1 \leq \sigma_2$ and $\sigma_1 < 1/4 - O(\varepsilon)$. Take $\ell = 6$ if $6\theta \leq 1 - \varepsilon/5$ and $\ell = 5$ otherwise; since $1/7 \leq \theta \leq 2/11$, the ℓ dyadic pieces of M_1^ℓ have lengths in $[M_1^\ell, 2^\ell M_1^\ell] \subset [X^{5/6-O(\varepsilon)}, X^{1-\varepsilon/10}]$. On $\mathcal{V}$, $|M_1^\ell|$ exceeds $(M_1^\ell)^{-\sigma_1}$, so one of the pieces exceeds $(M_1^\ell)^{-\sigma_1-o(1)}$. Lemma 4.3 with $\eta_0 = 5/6$ and $3\varepsilon/2$ in place of ε gives $|\mathcal{V}| \ll X^{2\sigma_1-9\varepsilon^2/16} \leq M_1^{2\sigma_1} M_2^{2\sigma_2} X^{-\varepsilon^2/4}$, because $M_1^{2\sigma_1} M_2^{2\sigma_2} \geq (M_1 M_2)^{2\sigma_1} = X^{2\sigma_1-o(1)}$.
- (b) $\sigma_2 < \sigma_1$ and $\sigma_2 < s(1-\theta) - O(\varepsilon)$. Apply Lemma 4.3 (with $3\varepsilon/2$ in place of ε) directly to M_2 , whose length is X^ϕ , with $\eta_0 = \max(9/11, \min(5/6, 1-\theta))$; in terms of θ the threshold is

$$s_2(\theta) := s(1-\theta) = \begin{cases} 1/4, & \theta \leq 1/6, \\ \frac{3-8\theta}{10(1-2\theta)}, & 1/6 \leq \theta \leq 2/11, \end{cases} \quad s_2\left(\frac{2}{11}\right) = \frac{17}{70}.$$

This gives $|\mathcal{V}| \ll X^{2\sigma_2-9\varepsilon^2/8}$, which suffices as in (a).

6.3. The sparse cases. If neither (a) nor (b) applies and the first branch fails, the second branch closes.

Proposition 6.1. *Assume (6) and that neither (a) nor (b) applies. Then (7) holds, provided ε is small in terms of ν .*

Proof. There are three regions.

Region A: $\sigma_1 \leq \sigma_2$, so $\sigma_1 \geq 1/4 - O(\varepsilon)$. By (8), $K = 2\sigma_1 + 5\theta(1-2\sigma_1) + 2(\sigma_2 - \sigma_1)(1-\theta) \geq \frac{1}{2} + 5u - O(\varepsilon)$.

Region B1: $\sigma_2 < \sigma_1$ and $\sigma_2 \geq 1/4 - O(\varepsilon)$. Since $\theta(1-2\sigma_2) \geq \theta(1-2\sigma_1)$, (8) gives $K \geq 5\theta(1-2\sigma_1) + 2\sigma_2 \geq \frac{1}{2} + 5u - O(\varepsilon)$.

In both regions $K > 1/a = 1 - u$ as soon as $6u > \frac{1}{2}$, i.e. $a > 12/11$; and $1925/1763 > 12/11$.

Region B2: $\sigma_2 < \sigma_1$, $\sigma_2 < 1/4 - O(\varepsilon)$ and $\sigma_2 \geq s_2(\theta) - O(\varepsilon)$. Then $\theta \geq 1/6$ (otherwise $s_2(\theta) = 1/4$), and $\sigma_2(2-4\theta) \geq \frac{3}{5} - \frac{8}{5}\theta - O(\varepsilon)$. By (8) and (6),

$$K \geq 4u + \theta + 2\sigma_2(1-\theta) - O(\varepsilon).$$

Multiplying by $1-2\theta > 0$,

$$\begin{aligned} (1-2\theta)(\theta + 2\sigma_2(1-\theta)) &\geq \theta(1-2\theta) + (1-\theta)\left(\frac{3}{5} - \frac{8}{5}\theta\right) - O(\varepsilon) \\ &= \frac{223}{385}(1-2\theta) + \frac{2}{385}(2-11\theta)(7\theta+2) - O(\varepsilon), \end{aligned}$$

and $(2-11\theta)(7\theta+2) \geq 0$ because $\theta \leq 2/11$. Hence $\theta + 2\sigma_2(1-\theta) \geq \frac{223}{385} - O(\varepsilon)$ and

$$K \geq 4u + \frac{223}{385} - O(\varepsilon) > 1 - u \iff 5u > \frac{162}{385} + O(\varepsilon) \iff \frac{1}{a} < \frac{1763}{1925} - O(\varepsilon),$$

which holds since $a \geq 1925/1763 + \nu$. $\square$

Combining (a), (b), Proposition 6.1 and the first sparse branch proves (4) for every bin, uniformly for $a \in [1925/1763 + \nu, 11/10]$.

Remark 6.2 (The extremal configuration). The constant $1925/1763$ is attained. At

$$\theta = \frac{2}{11}, \quad \sigma_2 = \frac{17}{70} = s_2\left(\frac{2}{11}\right), \quad \sigma_1 = \frac{517}{1925}, \quad a = \frac{1925}{1763},$$

the first branch is exactly saturated, $\theta(1-2\sigma_1) = \frac{162}{1925} = u$, and so is the second:

$$(5-8\sigma_1)\theta + 2\sigma_2(1-\theta) = \frac{998}{1925} + \frac{153}{385} = \frac{1763}{1925} = \frac{1}{a}.$$

Neither density case applies there ($\sigma_2 < \sigma_1$ and σ_2 is exactly at the threshold of M_2). If the long factor could also use the threshold $1/4$ at length $X^{9/11}$, Region B2 would disappear and the argument would give $a > 12/11$, i.e. $c > 23/11$.

7. COMPLETION OF THE PROOF

With (4) in hand, the remainder is the argument of [6, Sections 2 and 5] with the parameters left free.

7.1. The parameterized reduction.

Proposition 7.1. *Fix $\nu > 0$ and $c \in (2, 21/10]$, and let $1925/1763 + \nu \leq a \leq \min(11/10, c - 1)$, $P_1 = (\log X)^a$, $h = (\log X)^c$ and $h_1 = X^{99/100}$. There is $\varepsilon_0 > 0$, depending only on ν and c , such that, uniformly in a ,*

$$\frac{1}{X} \int_X^{2X} \left| \frac{1}{h} \sum_{\substack{x < pn \leq x+h \\ p \sim P_1}} \rho^-(n) - \frac{1}{h_1} \sum_{\substack{x < pn \leq x+h_1 \\ p \sim P_1}} \rho^-(n) \right|^2 dx \ll (\log X)^{-2-\varepsilon_0}.$$

Proof. By the Perron/Parseval reduction [6, Lemma 3.1] (whose proof is uniform in a and c) it suffices to prove

$$\int_{X^{1/1000}}^T |P_1(1+it)P(1+it)|^2 dt \ll \frac{T}{X/h} (\log X)^{-2-\varepsilon_0} \quad (T \geq X/h).$$

For $T \geq X$ this follows from the mean value theorem and the fact that each n has at most one prime factor $p \sim P_1$. On the complement of $\mathcal{U}$ in $[X^{1/1000}, X]$ one uses $|P_1(1+it)| < P_1^{-\varepsilon/10}$, the improved mean value theorem [6, Lemma 3.3] and the rough-number sieve bound of [6, Section 5]. On $\mathcal{U}$ one uses the decomposition [6, Proposition 2.2]. The type I and type I/II components are bounded as in [6, Sections 5.1, 5.2]: the power savings $X^{-(1-1/a)/20+O(\varepsilon)}$ and $X^{-\varepsilon(1-1/a)/4+\varepsilon/100+O(\varepsilon^2)}$ left there are genuine because

$$\frac{1-1/a}{4} \geq \frac{1-1763/1925}{4} = \frac{81}{3850} > \frac{1}{100}.$$

The type II components are bounded by Proposition 5.1 on $\mathcal{U}_1$, trivially on $\mathcal{U}_3$, and by (4), Section 6, on the $O((\log X)^2)$ bins of $\mathcal{U}_2$. $\square$

7.2. From variance to exact semiprimes. Fix $c \in (3688/1763, 21/10]$ and $\Delta > 0$ with $c - 1 - \Delta > 1925/1763$, and take the disjoint dyadic prime blocks $p \sim P_1$ with $(\log X)^{c-1-3\Delta/4} < P_1 < 2P_1 \leq (\log X)^{c-1-5\Delta/8}$. For each block, $a = \log P_1 / \log \log X$ lies in $[1925/1763 + \nu, 11/10]$ for some $\nu = \nu(c, \Delta) > 0$, so Proposition 7.1 applies uniformly. Summing over the $\asymp \log \log X$ blocks with Cauchy–Schwarz, the long-interval lower bound [6, Theorem 2.1(ii)], Chebyshev’s inequality and $\rho^- \leq \mathbf{1}_{\mathbb{P}}$ give, for all $x \in [X, 2X]$ outside a set of measure $O(X/(\log X)^\delta)$,

$$\#\{pn \in (x, x+h] : p, n \text{ prime}, p \text{ in a block}\} \gg_{c,\Delta} (\log X)^{c-1}.$$

Three routine passages complete the proof of Theorem 1.3 for $c \leq 21/10$: from real to integer x (using the same bound for $h/2$); from $[X, 2X]$ to $[1, X]$ (dyadic summation preserves the exponent δ); and from the blocks to the window

$$((\log x)^{c-1-\Delta}, (\log x)^{c-1-\Delta/2}].$$

For $c > 21/10$ the existence statement follows from the case $c = 21/10$.

7.3. **The value 2.092.** Take $c = 2.092 = 523/250$ and $\Delta = 10^{-4}$. Then

$$c - 1 - \Delta = 1.0919 = \frac{10919}{10000}, \quad \frac{10919}{10000} - \frac{1925}{1763} = \frac{197}{17630000} > 0,$$

so Theorem 1.3 applies. Its prime window satisfies

$$((\log x)^{1.0919}, (\log x)^{1.09195}] \subset ((\log x)^{1.091}, (\log x)^{1.092}],$$

and its lower bound is $\gg (\log x)^{1.092}$. This proves Theorem 1.2.

8. THE LEAN FORMALIZATION

The accompanying Lean 4 project (Lean 4.34.0, Mathlib v4.34.0) proves

```
theorem ExactSemiprimes.mainTheorem_exponent_2092
```

```
  (inputs : ExternalInputs) (ext : ParameterExtensionInputs) :
    Results.CleanTheoremStatement      -- Theorem 1.2
theorem ExactSemiprimes.mainTheorem_parameter
  (inputs : ExternalInputs) (ext : ParameterExtensionInputs) :
    Results.ParameterTheoremStatement -- Theorem 1.3
```

- **ExternalInputs** collects published theorems, each transcribed in its printed parameter range: Guth–Maynard’s Theorem 1.1 and Proposition 12.1, Heath-Brown’s sparse mean value theorem, Hildebrand–Tenenbaum’s smooth-number bound, the decomposition, equations and lemmas of [6] that are used verbatim, the mean value theorem of Iwaniec–Kowalski, and a few others.
- **ParameterExtensionInputs** collects five labelled statements that are printed only at $c = 2.1$ (or $a \in [1.0999, 1.1]$) and whose printed proofs are uniform in the parameters: the Parseval/Perron reduction [5, Lemma 14], [7, Lemma 1], [6, Lemma 3.1]; the long-interval lower bound [6, Theorem 2.1(ii)]; the rough-pair sieve bound of [6, Section 5]; and the type I and type I/II bounds of [6, Sections 5.1, 5.2] for $a \in [1925/1763, 11/10]$.
- Everything else is proved: Lemma 4.3 (**Hybrid/RefinedDensity.lean**), Proposition 5.1, the two sparse branches, the three cases of Section 6 (**Final/TypeIIDensityBins.lean**, **Final/TypeIIComponent.lean**), the Perron target, the variance and counting arguments and the passage to Theorems 1.2 and 1.3.
- There is no `sorry` and no project `axiom`. The command `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

The file `ASSUMPTIONS.md` of the project lists every hypothesis with its source.

ACKNOWLEDGEMENTS

Siddharth Iyer was used as a “meat proxy”¹ for this project. He collected the outputs of the language models and told ChatGPT to discover problems in the work of Matomäki. The project began with the following prompt, an important human idea, which he gave to ChatGPT:

Identify works of Matomaki and detail five problems from Matomaki’s works that seem within reach, they could be 1) an improvement of a result of Matomaki, 2) Answering a question of Matomaki, 3) Disproving a question of Matomaki. For now you do not need to supply proofs for five such problems, operate on vibes and slight rigour, also try and balance it so that the problems are “interesting” to the broader mathematical community. For each such problem also detail a reference that relates to Matomakis work. Detail the five problems in a md file, so that I can supply it towards future agent runs (that actually attempt to solve some of the problems).

¹A *meat proxy* is a human who serves as the physical go-between for AI systems, much as a network proxy forwards traffic: typing or pasting prompts in, carrying the outputs from one model or tool to another, moving files around, and running whatever the systems cannot run for themselves.

REFERENCES

- [1] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, Ann. of Math. (2) **203** (2026), 623–675; [arXiv:2405.20552](#).
- [2] D. R. Heath-Brown, *The differences between consecutive smooth numbers*, Acta Arith. **184** (2018), 267–285; [arXiv:1808.02947](#).
- [3] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.
- [4] H. Iwaniec and E. Kowalski, *Analytic number theory*, Amer. Math. Soc. Colloq. Publ. **53**, 2004.
- [5] K. Matomäki and M. Radziwiłł, *Multiplicative functions in short intervals*, Ann. of Math. (2) **183** (2016), 1015–1056.
- [6] K. Matomäki and J. Teräväinen, *Almost primes in almost all short intervals II*, Trans. Amer. Math. Soc. **376** (2023), 5433–5459; [arXiv:2207.05038](#).
- [7] J. Teräväinen, *Almost primes in almost all short intervals*, Math. Proc. Cambridge Philos. Soc. **161** (2016), 247–281.